Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/03/2020
PI Name:	Main, Bob M.S.		
Project Title:	Create the Technical Requirements, Strategic Roadmap, and Financial Requirements to Develop a Multifunction Ophthalmic Device for Comprehensive Ophthalmic Vision Testing on Deep Space Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	84009	**Congressional District:**	4
Comments:			
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COI Name (Institution):	Buscemi, Philip O.D. (Buscemi Group, LLC.)		
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	Focused Investigation Project Primary deliverables: Create a panel (Vision for Mars Committee) of ophthalmic instrument design/manufacturing experts to review the project needs and help determine the technology roadmap/budget. The Principal Investigator (PI) will meet with the panel throughout this project and conduct several web meetings to get their input. The PI will solicit candidates for the panel from NASA, Translational Research Institute for Space Health (TRISH), and ophthalmic industry consultants. Conduct meetings with key players that have knowledge of what the device is expected to do. Meeting/focus-groups with key NASA and TRISH personnel and former astronauts will be conducted to get the user perspective. Some of the meetings will be with: NASA Vision Team = Capt Tyson Brunstetter, - NASA Aerospace-Research Optometrist / Deputy Spaceflight Associated Neuro-ocular Syndrome (SANS) Clinical Lead and William Tarver, MD, MPH - Chief, Clinical Services NASA Johnson Space Center; members of the Human Research		

Task Description:	Program, and others within NASA that they recommend. Work with Dr. Dorit Donoviel (Director of TRISH) to develop a list of ophthalmic experts and former astronauts associated with TRISH/Baylor College of Medicine to interview. Web Vision's global network of ophthalmic/optical industry experts that monitor technology development are a good source to find out what companies are working on technology that might be useful for the Multifunctional Ophthalmic Device (MOD). Conduct meetings with technology companies/industry consultants and go to technology conferences. Learn what eye health technology is currently available or in development that meets initial device specifications. There are three primary conferences that most of the key optical industry experts attend. Additionally, most of the ophthalmic companies go to these same conferences and we will be able to meet with their Business Development Executives (most we already know and have met with previously). The three key conferences are: 1) American Academy of Ophthalmology, 2) The Optical Industry Summit, 3) European Association for Vision and Eye Research. Develop and document detailed device specifications based on NASA/TRISH and industry feedback. Get agreement and signoff of the specifications from the NASA Vision Team. Create a detailed technology strategy/roadmap on how the device could be developed and what kind of technology it will take to develop it. Find out what technology is currently available and what needs to be developed/invented. Develop an approximate budget of what it will take to develop this MOD technology. Through digital channels, educate the industry what the project is about, how to provide input, provide updates, and show how anyone can follow and participate in this project process. Conduct a final meeting with NASA and TRISH representatives at the end of the project to present the findings. Discuss the team's assessment of what is possible to develop, what technology is currently available, what needs to be created/invented, and what the overall projected budget will be to develop the device. Our team will provide written documentation that will include the information necessary for NASA/TRISH to have the MOD technology developed in future projects/grants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Problem That Needs to Be Solved: Approx. 63% of long-duration crewmembers present with one or more signs of Spaceflight Associated Neuro-ocular Syndrome (SANS), including optic disc edema, globe flattening, choroidal folds, cotton wool spots, refractive shifts, etc. SANS is a top risk and priority to the International Space Station (ISS) Program, Human Research Program (TA06) technology priority, and Human System Risk Board (rated at 3x4 for Deep Space Journey). NASA is currently conducting vision tests on the ISS using COTS (Commercially Available Off the Shelf) eye health evaluation equipment. To properly evaluate and monitor the astronauts eye health, NASA currently needs several different vision testing devices on the ISS, such as retinal imaging, optical coherence tomography (OCT), visual fields, and visual acuity. Unfortunately, this equipment was not designed for the harsh elements of space (i.e., radiation hardness), is complicated to operate, and is too large for extended deep space exploration and spacecraft that will be smaller than the ISS. In order to conduct the required vision tests in smaller spacecraft, NASA is looking for the development of a smaller, compact, single user device that would allow them to continue to monitor the astronauts eye health during extended space exploration missions. How This MOD will Solve the Problem: This Multifunctional Ophthalmic Device (MOD) will be designed to include all of the required eye health testing equipment in a compact device, and as a result it will save significant space, astronaut time (due to the self-testing feature), and it will be designed to handle the harsh environment of space. NASA needs a MOD type of device to monitor/detect/treat eye health related problems associated with SANS. Currently, no ultra-compact (all-in-one) COTS devices are available and to our knowledge no company is currently working on such a device.
Task Progress:	Original Project Aims/Objectives: This project is part of an overall Strategy/Plan to develop a compact, comprehensive Multifunctional Ophthalmic (all-in-one) vision testing Device (MOD) to be used during extended space missions to deliver a comprehensive eye health assessment of the astronauts. This MOD will include vision testing functionalities such as retinal imaging, visual-fields, OCT (Optical Coherence Tomography), and Visual Acuity assessment, all inside of one compact device. This project will accomplish the following: Project Primary Deliverables (Phase 1): 1. Develop Detailed Specifications for MOD; 2. Determine Best Strategy - Reach Out to Global Ophthalmic Community; 3. Develop a Technology Roadmap for Building the Device-- a) Potential it can be done, b) How to do it (and with whom), c) Budget, d) Timing Background -- NASA has recognized that Commercial Off-The-Shelf (COTS) vision testing/diagnostic equipment, individually and collectively, are too large for smaller future deep space spacecraft. Additionally, COTS were not designed for the rigors of space (including electronics that have not been radiation hardened), are skill-dependent (making it difficult for astronauts to use), and are time consuming. NASA has asked the Vision for Mars team to investigate the development of the MOD that have these general requirements: * Single user (one astronaut to operate), * Very easy to operate (minimal training -- possibly computer voice directed), * Small footprint (small as possible/practical) -- in one single device if possible, * Lightweight (under 15lbs./6.8kg.), * Electronics protected from radiation, * Easy to fix in-flight (electronic boards can be easily replaced with spare boards). In order of priority, these are the vision testing capabilities NASA requested: 1. OCT (Optical Coherent Tomography), 2. Retinal imaging, 3. Perimetry (Humphrey Compatible), 4. Optical Biometry, 5. Objective refraction or difference in refraction from baseline, 6. Amsler Grid or more sensitive measurement of metamorphosia (such as hyper-Acuity), 7. Color vision testing. Project Summary of Key Findings and Results (Phase 1) #1 Deliverable -- Develop Detailed Specifications for MOD. We worked with NASA and industry experts to develop comprehensive specifications for the MOD. The specifications have been completed (as much as possible given that NASA has not developed all of the relevant specifications for the deep space vehicles yet). #2 Deliverable -- Determine Best Strategy and Reach Out to Ophthalmic Community. A. We met with over 30+ companies to find out what relevant technology is available to help develop the MOD. B. We created a Vision for Mars Executive Committee made up of high-level executives from several ophthalmic companies and industry experts to help determine the best strategy to develop the MOD. C. Additionally, we did a global search to find a technology development partner that could assist us in designing the MOD and build a mock-up model of the proposed MOD device. We selected a UK based company called TTP plc (https://). #3 Deliverable -- Develop a MOD Technology Roadmap. After

	investigating several potential designs, we selected the following: A single compact device that has two different optical paths with two eyepieces. One optical path that has all the narrow angle devices (OCT, retinal imaging, aberrometry) and another optical path that has the wide-angle device (visual field). 1. Two separate optical paths: a. Narrow-angle path that contains: i. OCT - optionally modified to include axial biometry, ii. Retinal imaging camera, iii. Aberrometry using Shack-Hartmann method, iv. Objective refraction; b. Wide-angle path: i. Visual field (Humphrey 24-2 compatible). 2. Amsler Grid / Color vision testing via laptop screen (using existing in-flight laptop). Summary of Proposed MOD Device: One Device with Two Separate Internal Optical Pathways and Two Separate Eyepieces, Self-Imaging, Projected Size - 10 X 10 X 10 (250 mm X 250 mm X 250 mm), Projected Weight = 13 lbs (6 kg), Power Supply for Device Does Not Come from Laptop, Power Will Need to Come from Station/Capsule; DC Power or Might Need Converter to AC - TBD.
Bibliography Type:	Description: (Last Updated:)